

SYNOPSIS



On 27 October 2010, two Rigid Inflatable Boats (RIBs) belonging to the Cardiff Bay Yacht Club (CBYC) collided while transporting a number of children across Cardiff Bay. As a result of the impact three children were thrown into the water, while others suffered abrasions and muscular injuries.

The children, all girls, were participants in an International Optimist Class Association (UK) (IOCA) training event. Following an evening social activity, the group were being ferried from the clubhouse across Cardiff Bay to their accommodation.

The two RIBs carried no navigation lights or torches and were proceeding at about 20 knots in the dark. They were being driven by 17 year olds who held Royal Yachting Association (RYA) Powerboat Level 2 certificates, a qualification that did not equip them for night-time operations.

A Cardiff Harbour Authority (CHA) 'common sense regulation' which stated that '*persons under the age of 18 years must not take charge of a power driven vessel with an engine exceeding 5 horse power*' was in force at the time of the accident. However, the CBYC chief instructor was unaware of the regulation when he appointed the drivers to the two RIBs, which were equipped with 50 horsepower engines.

The investigation identified a number of safety issues. These include a lack of planning and risk assessment for the night-time ferrying activity, poor awareness of harbour regulations, and shortcomings in the CBYC's maintenance and safety procedures for its boats.

A recommendation has been made to the Cardiff Bay Yacht Club which seeks to improve the club's safety management system and control of risk. Recommendations have also been made to the Cardiff Harbour Authority to confirm the appointment of a designated person and measures to improve stakeholders' awareness about the statutory role of the authority

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF CARDIFF BAY YACHT CLUB RIBS AND ACCIDENT

Vessel details

RIB 1	:	Ribcraft 4.8 Pro
Built	:	2010
Construction	:	GRP hull with rubber inflatable tubing
Length overall	:	4.8m
Engine power and type	:	50hp, Suzuki Four Stroke
Service speed	:	20 knots

RIB 2 Tornado Multi Purpose 5.4

Built	:	1997
Construction	:	GRP with rubber inflatable tubing
Length overall	:	5.4m
Engine power and type	:	50hp, Yamaha Four Stroke
Service speed	:	20 knots

Accident details

Type of marine casualty or incident	:	Serious Marine Casualty
Time and date	:	2008 on 27 October 2010
Location of incident	:	Cardiff Bay, approximately 250m north-east of Cardiff Bay Yacht Club
Persons on board	:	RIB 1: 7 (driver + 6 girl students) RIB 2: 6 (driver + 1 adult + 4 girl students)
Injuries	:	One girl suffered injuries which were later diagnosed as a "traumatic brain injury of moderate severity" and post-concussion syndrome; another suffered cuts, bruising and torn rib muscles; and a third suffered whiplash and injuries to her thoracic and lower spinal regions.

1.2 BACKGROUND

The role of the International Optimist Class Association (UK) (IOCA) is to govern and promote the sailing and racing of Optimist¹ dinghies and to encourage and facilitate training of young people to sail and race. IOCA is run on a voluntary basis with funding support for some activities, coaching and international team squad training, provided by the Royal Yachting Association (RYA).

IOCA organises an annual programme of yacht racing and race coaching events for its members, and runs four major competitive events annually in accordance with RYA Junior Racing policies². The participants in these events are typically members of the national and intermediate sailing squads. The teams which represent the UK in international sailing competitions are selected from these squads.

IOCA normally undertakes thorough, documented, risk assessments and provides the race officers, safety boats and competent drivers for these events.

In addition to the structured training and racing events, an open race coaching week, specifically for girls, is held annually in October. During the summer of 2010, the IOCA committee member responsible for girls' training made arrangements with the chief instructor of the Cardiff Bay Yacht Club (CBYC) to hold the 2010 girls' training week in Cardiff Bay. The chief instructor was an RYA qualified sailing instructor, who had organised several similar IOCA events at different venues in the past.

This was the first time the event had been held in Cardiff; previously it had been staged at a sailing club with on-site accommodation for the participants.

The CBYC has no on-site accommodation and, therefore, rooms were booked for the participants at a suitable youth hostel near Mermaid Quay on the opposite side of Cardiff Bay from the yacht club (**Figure 1**). It was agreed by the chief instructor that the girls would be ferried daily between the hostel and CBYC in club boats.

No risk assessments were undertaken by either IOCA or the chief instructor in respect of the event or the journeys across the bay in CBYC boats.

The event was held in the school half-term with 24 girls aged between 10 and 14 taking part. Their sailing abilities varied between racing novices and members of the Optimist national and intermediate squads.

1.3 NARRATIVE

1.3.1 Joining instructions

On 19 October 2010, IOCA informed the participants' parents by email that the girls should arrive at CBYC by 1000 on 25 October and that the event would finish at 1600 on 28 October.

The joining instructions gave information regarding accommodation and meals, and provided advice about clothing requirements for both sailing and the planned evening activities, which included swimming and ice skating.

¹ The Optimist sailing dinghy is 2.36m in length and is the entry level dinghy for junior sailing (for sailors up to 15 years of age).

² The RYA publishes a range of policy guidance for junior sailing which can be viewed at: <http://www.rya.org.uk/racing/youthjunior/information/Pages/Policies.aspx>

Figure 1



Cardiff Bay

IOCA made arrangements for three of the parents to stay with the girls for the duration of the event to act as “house mothers”.

1.3.2 Training event

The girls arrived at CBYC on the morning of Monday 25 October and were advised of the programme for the training event. They were divided into four groups, each supervised by an RYA sailing coach, and then went afloat to commence their training.

The first day's sailing completed at about 1600 and the girls then changed into their leisure clothing prior to their evening activity, which on this night was swimming in a pool close to the yacht club. They returned to the club at about 1800, where they had their evening meal. It was dark by the time the girls were taken from CBYC to Mermaid Quay (**Figure 1**) in four RIBs owned by the club.

For the passage to Mermaid Quay, the girls all wore the buoyancy aids that they had used while sailing during the day. The majority of these buoyancy aids were dark coloured and none had lights attached to them. During the trip, the girls predominantly sat on the tubes of the RIBs, as the boats had insufficient seats to accommodate all the passengers.

The following morning the girls, together with two of the house mothers, were collected from Mermaid Quay and returned to the yacht club in the four RIBs. The chief instructor watched the boats as they crossed the bay and saw some of the

RIBs being driven in an apparently reckless manner. He later reprimanded the relevant drivers, informing them that such behaviour was not acceptable and should not be repeated.

The previous evening, the IOCA committee member who was organising the event had noticed that none of the club's RIBs carried navigation lights. She discussed this with the chief instructor during the morning, and he advised her that the boats did not need to show navigation lights.

On completion of the second day's sailing the girls had their evening meal at the club and were taken back to Mermaid Quay, again in the four RIBs, where they arrived at about 1730. They returned to the hostel and then went ten-pin bowling. Several of the coaches also attended this activity and returned across the bay to CBYC in the club RIBs later that evening.

On Wednesday 27 October at about 0900, the girls and two of the house mothers were collected from Mermaid Quay by the four RIBs and taken across the bay. One or two of the RIBs were occasionally swerving, so as to thrill the occupants, some of whom appeared to encourage this action. Two of the house mothers were quite concerned about this behaviour, but did not consider they could comment to the drivers on the operation of the boats.

The day was divided between group training sessions, debriefs ashore, lunch and then races in the afternoon. The girls finished sailing at 1630 and changed into leisure clothing they had brought with them in the morning, for the evening activity of ice skating.

The ice rink was located close to the yacht club and the girls were there from about 1700 to 1830, after which they returned to the club for their evening meal. At about 1950 the girls finished their meal and packed their sailing kit into bags to take back with them to the hostel.

During the ice skating, one of the girls had fallen heavily and had hurt her back so she, together with three other girls and two of the house mothers, returned to the hostel by car.

1.3.3 The collision

At about 2000 the remaining 20 girls donned their buoyancy aids and boarded the four RIBs for the trip to Mermaid Quay.

The chief instructor was driving the lead RIB, and the second RIB was driven by a sailing instructor who held an RYA advanced powerboat certificate. Both drivers were carrying very high frequency (VHF) radios. The two remaining RIBs were driven by two 17 year old sailing instructors who did not have VHF radios with them.

In the darkness, the four RIBs left the pontoons and proceeded at slow speed through the mooring area adjacent to the club. Once clear, the first two RIBs increased speed and headed directly towards Mermaid Quay.

The RIBs driven by the two 17 year olds were behind the leading boats when they cleared the mooring area. As they entered clear water, the last two boats increased speed to about 20 knots and started to separate. One, a dark blue tubed Ribcraft

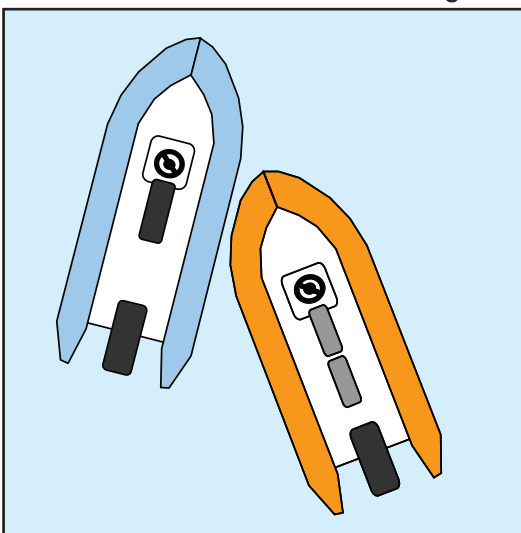
RIB, with its driver and six girls on board, set off to the left of the track taken by the first two RIBs. The other, an orange tubed Tornado RIB, with its driver, a house mother and four girls on board, set off to the right, passing close to moored craft on the outer edge of the moorings (**Figure 2**). The driver of this RIB soon became concerned about approaching too close to the locks, and swung the boat back to port towards the centre of the bay.

Figure 2



Track of the RIBs prior to the collision

Figure 3



Angle of RIBs at time of collision

The Tornado RIB was probably heading towards the mouth of the River Taff when the driver suddenly saw the starboard side of the Ribcraft RIB very close ahead. Although the driver reduced the boat's speed and started to turn to starboard, collision could not be avoided (**Figure 3**).

The driver of the Ribcraft RIB had not seen the other boat before the collision occurred. On impact, the Ribcraft RIB heeled over to port and the port buoyancy tube of the Tornado RIB momentarily went under the starboard buoyancy tube of the Ribcraft RIB before the boats separated.

1.10 REPORTING THE ACCIDENT

The RYA became aware of the accident on the morning of 28 October, and later that day received an interim report from CBYC's chief instructor. A more comprehensive report was prepared by members of CBYC and sent to the RYA on 24 November 2010. The accident was reported to MAIB on 25 November 2010, and to the harbourmaster on 14 December 2010.

1.11 PREVIOUS ACCIDENTS

In August 2008⁸, a female passenger on board a 9m RIB suffered a lower back wedge compression fracture. The boat was conducting a 'thrill ride' in the Bristol Channel and the injury occurred when the passenger landed heavily on her seat after she had been momentarily lifted into the air due to the motion of the craft.

In February 2009, a RIB operating from the Plas Menai National Watersports Centre in Wales capsized while carrying a party of school children, injuring one of them. At the time of the accident the children had completed the main activity of the day and were being taken from the centre for an evening trip to go kite flying.

The MAIB conducted an investigation into the accident and published a report⁹ which concluded that the RIB had capsized when it encountered rough water during an unauthorised deviation from a planned powerboat trip. The investigation identified a number of safety issues, which included shortcomings in planning, control, leadership and communications.

As a result of this accident the MAIB made a recommendation to the RYA to produce guidance for RYA training centres offering powerboating activities other than RYA recognised tuition (**Annex E**). The report made recommendations to the Plas Menai Centre to review its risk assessments and to provide staff with guidance on, inter alia, considerations for 'journeying'¹⁰ and similar activities.

Some of the CBYC members who undertook the club's investigation of the collision were aware of the Plas Menai accident but were not aware of the MAIB investigation or the RYA guidance to training centres.

In January 2011 the MAIB issued a report¹¹ following the investigation of an accident in which a passenger travelling in a RIB received serious spinal injuries while not seated in an appropriate seat.

⁸ http://www.maib.gov.uk/publications/investigation_reports/2009/celtic_pioneer.cfm

⁹ http://www.maib.gov.uk/publications/investigation_reports/2009/rib_6.cfm

¹⁰ Journeying: the term given when RIBs or other boats at a training centre are used as a means of transport rather than for powerboat training or tuition.

¹¹ http://www.maib.gov.uk/publications/investigation_reports/2011/delta_rib.cfm

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 PLANNING OF THE EVENT

2.2.1 IOCA

The girls' open race training event, although a regular feature of the IOCA calendar, was not subject to the rigorous planning and risk assessments undertaken by IOCA for race meetings and other major sailing events.

The arrangements for the race training week, including the transport of the girls across Cardiff Bay were, in the main, left to CBYC's chief instructor to organise. This led to IOCA having a lack of oversight of the event, even though the participants and their families could reasonably have expected that it would be subject to the same high standards of safety apparent at the other IOCA events.

There is no evidence to suggest that the race training, which took place during the day, was not conducted safely. However, the fact that IOCA was not involved in assessing the risks involved in transporting the children across Cardiff Bay at night was a significant omission in the planning of this event.

2.2.2 CBYC chief instructor

When the CBYC chief instructor agreed to stage the open race training event in Cardiff Bay, hosted by CBYC, he was bringing the event to waters he knew intimately.

He had sailed the waters of Cardiff Bay for many years and, as chief instructor at CBYC, had organised and run sailing events from the club in the past. This familiarity might have led to a degree of complacency regarding the hazards involved, and thus the need to undertake a specific risk assessment for this event was not recognised.

This was the first occasion on which, as part of a sailing event based at CBYC, participants were to be transported across the bay at night. At no stage was the planning for this part of the event given the consideration required for such a potentially hazardous passage.

2.2.3 Selection of the RIB drivers

From the initial planning discussions for this event, it was known that there would be a requirement to operate power-driven craft at night to ferry the girls between CBYC and their accommodation.

The syllabus of PB2 does not include powerboat operation by night and the CHA 'common sense regulations' for Cardiff Bay did not permit anyone under the age of 18 to operate power-driven craft with engines of more than 5hp at any time.

The CBYC chief instructor should have recognised that the two 17 year old coaches were not sufficiently trained or experienced to safely operate power-driven craft at night without close supervision. He should also have been aware of the harbour authority's stance, ie that they should not operate the club RIBs in Cardiff Bay at any time.

If an appropriate risk assessment had been undertaken for the evening trips, the unsuitability of the young drivers for the task would have been recognised.

2.3 THE ROLE OF VOLUNTEERS

Participation in sailing events like the IOCA training week is undoubtedly beneficial to children, both for the improvement of their sailing skills and for the development of their self-confidence and resilience. However, when involved in such events, adult volunteers and coaches must ensure that they remain vigilant at all times to the potential risks to the children.

During the week, some of the RIBs were observed being driven inappropriately, apparently to thrill the children at their request, when journeying between Mermaid Quay and the club. Although house mothers were present on the boats when this activity took place and were concerned, they did not indicate to the drivers that this was inappropriate behaviour. This was probably due to uncertainty as to whether it was appropriate for them to question or challenge aspects of what they considered to be an inappropriate activity.

Organisers of events involving volunteers should ensure that the volunteers are provided with clear guidelines as to their responsibilities to raise any concerns they have about the conduct of activities, and the means by which they should do so.

2.4 THRILL SEEKING BEHAVIOUR

Driving RIBs at speed through waves and across the wash of other craft, known commercially as "thrill rides", can be carried out safely providing all the risks are properly identified and mitigated.

The MAIB reports listed in Section 1.11 all concern accidents where an activity was not well considered, or the injured passenger was not appropriately seated, or both. In two of the cases the passengers suffered serious spinal injuries. The safety issues arising from these investigations highlight the importance of careful risk consideration prior to the trip, proper driver training, and the necessity for appropriate passenger seating.

At an early stage of the IOCA event some of the RIB drivers, including one of the drivers involved in the collision, had been seen to drive in such a way as to make the ride more exciting for their passengers. These drivers did not have the training or experience required to appreciate the potential dangers of such behaviour.

The CBYC chief instructor reprimanded the drivers involved in "thrill seeking behaviour", but did not take the opportunity to review their suitability to be in charge of the boats while ferrying the girls to their accommodation at night.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT WHICH HAVE RESULTED IN RECOMMENDATIONS

1. The CBYC chief instructor should have recognised that the two 17 year old coaches were not sufficiently trained or experienced to safely operate power-driven craft at night without close supervision. He should also have been aware that the CHA had published a 'common sense regulation' which directed that they should not operate high-powered vessels such as the club RIBs in Cardiff Bay at any time. [2.2.3]
2. The guidance issued by the RYA to its training centres following the Plas Menai accident (**Annex E**) included advice that boat passengers should, where appropriate, be provided with a seat. CBYC should ensure that the recommended maximum capacity of its boats as marked on the builder's plate is not exceeded. It should also provide guidance on the use of its boats for the carriage of persons in excess of the available seating. [2.7]
3. CBYC should review its system for the care and maintenance of club boats to ensure they are always in a good and serviceable condition. [2.10]

3.2 OTHER SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION ALSO LEADING TO RECOMMENDATIONS

1. The CBYC chief instructor reprimanded the drivers involved in thrill seeking behaviour but did not take the opportunity to review their suitability to be in charge of the boats. [2.4].
2. With no designated person appointed, the duty holder is not complying with the requirements of the PMSC and is denying itself a source of independent assurance and advice. Cardiff City Council should consider the appointment of a designated person as soon as is practicable. [2.12.2]
3. The lack of knowledge of essential marine safety information demonstrated by the chief instructor and other senior members of CBYC, a major stakeholder, suggests that there is insufficient understanding of the CHA's roles and responsibilities among some bay users. CHA should take further measures to clarify its regulations and to raise its profile both afloat and ashore to ensure that harbour users understand and respect the role of the statutory harbour authority in maintaining a safe marine environment in Cardiff Bay. [2.12.3]
4. It is regrettable that the lessons from a previous, similar accident were not apparently recognised or understood by senior members of CBYC. The club should ensure that the safety lessons from previous accidents and RYA guidance, particularly in relation to risk assessments, are disseminated to, and properly understood and put into practice by all its members. [2.13]

3.3 SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION WHICH HAVE BEEN ADDRESSED OR HAVE NOT RESULTED IN RECOMMENDATIONS

1. There is no evidence to suggest that the race training, which took place during the day, was not conducted safely. However, the fact that IOCA was not involved in assessing the risks involved in transporting the children across Cardiff Bay at night was a significant omission in the planning of this event. [2.2.1]
2. This was the first occasion on which, as part of a sailing event based at Cardiff Bay Yacht Club, participants were to be transported across the bay at night. At no stage was the planning for this part of the event given the consideration required for such a potentially hazardous passage. [2.2.2]
3. Organisers of events which require the use of volunteers should ensure that the volunteers are provided with clear guidelines as to their responsibilities to raise any concerns they have about the conduct of activities, and the means by which they should do so. [2.3]
4. Good local knowledge of a particular area does not obviate the fundamental requirement to comply with the COLREGS. That the CBYC chief instructor was content to allow high speed power-driven craft to operate at night, without navigation lights, demonstrates either a basic lack of knowledge and understanding of the COLREGS, or a failure to recognise the risks associated with four high speed RIBs crossing the bay in darkness in an uncoordinated manner. [2.5]
5. The boats involved in the collision were being driven by drivers who had not achieved a relevant qualification or gained experience in night navigation. They were being driven at speed and without lights, in an area where flotsam was a known hazard and with passengers sitting on the buoyancy tubes. Proper consideration of the night passage across the bay could have identified these hazards, and appropriate control measures, including transiting at a safe speed, could have been established. [2.5, 2.6, 2.7]
6. The transportation of the participants across Cardiff Bay at night should have been properly assessed in advance of the event. The supply of suitable lifejackets, fitted with lights and whistles, for these journeys should have been considered as an essential control measure. [2.8]
7. To avoid recurrence, the immediate care and treatment offered to children involved in high impact accidents should be considered as part of the event risk assessment. The requirement to seek prompt medical attention should be included in the event's contingency measures. [2.9]
8. The carriage of safety equipment, maintained in a good condition and stowed in appropriate containers is a prerequisite for the operation of a safety boat. The fact that this equipment was not available for the IOCA training week indicates that the club should review its safety management system in relation to the operation and maintenance of its boats. [2.10.2]
9. CBYC should review the provision and use of radios, with the aim of ensuring that all its water-borne activities have appropriate means of raising the alarm and of communicating during emergency situations. [2.10.3]

10. The RYA should ensure its inspectors complete annual inspection checklists proficiently, highlighting items requiring attention together with the responsibility and time scale for the recommended remedial action. [2.11]
11. The lack of any formal review of the authority's safety management system for almost 9 years is contrary to the requirements of the PMSC, and the duty holder should take action to ensure that this situation is not repeated in the future. [2.12.1]